

# Heteroskedasticity of Known and Unknown Form

Teaching note

Aboody, Barth, and Kasznik (1999), *Journal of Accounting and Economics*

ACCTG 597A · Fall 2026 · Sam Bonsall · Penn State

## ABK correct for heteroskedasticity twice

*“We deflate both  $\Delta OPINC$  variables and  $REV_t$  by beginning of year  $t$  market value of equity.”*

Aboody, Barth, and Kasznik (1999, p. 160)

*“Reported  $t$ -statistics are based on White (1980) standard errors (Barth and Kallapur, 1996), and the estimation controls for outliers using a robust regression technique.”*

Aboody, Barth, and Kasznik (1999, p. 162)

- Equation (1) is deflated by market value and then given White standard errors; this note asks what each correction assumes and when each is the appropriate one.

## Equation (1) is a levels model over MVE

As printed (p. 158);  $\Delta OPINC$  and  $REV$  are deflated by beginning market value of equity,  $MVE$ :

$$\Delta OPINC_{t+\tau,i} = \sum_{Y=83}^{95} \alpha_{0Y} YR_{Yti} + \alpha_1 REV_{ti} + \alpha_2 \Delta OPINC_{ti} + \alpha_3 MB_{ti} + \alpha_4 ASSETS_{ti} + \varepsilon_{ti} \quad (1)$$

In pounds (levels, controls omitted):

$$\Delta OPINC_{t+\tau} = \alpha_{0Y} MVE + \alpha_1 REV + u, \quad \text{Var}(u_i) = \sigma_i^2, \text{ presumably increasing in size.}$$

As estimated:  $\Delta OPINC_{t+\tau}/MVE = \alpha_{0Y} + \alpha_1 (REV/MVE) + u/MVE$ ; the year intercepts are the levels coefficients on  $MVE$ .

- Dividing by market value is a statement about the error variance, not only about units.

## Heteroskedasticity misstates the variance

Model  $y_i = \alpha + \beta x_i + u_i$ ,  $\text{Var}(u_i | x) = \sigma_i^2$ ;  $\tilde{x}_i$ :  $x_i$  partialled on the other regressors;  
 $S_{xx} = \sum_i \tilde{x}_i^2$ .

$$b = \beta + \frac{\sum_i \tilde{x}_i u_i}{S_{xx}} \quad \Rightarrow \quad \text{Var}(b | x) = \frac{\sum_i \tilde{x}_i^2 \sigma_i^2}{S_{xx}^2}$$

The conventional  $s^2/S_{xx}$  targets  $\text{Var}_{\text{conv}} \equiv \bar{\sigma}^2/S_{xx}$ ,  $\bar{\sigma}^2 = n^{-1} \sum_i \sigma_i^2$ ; dividing,

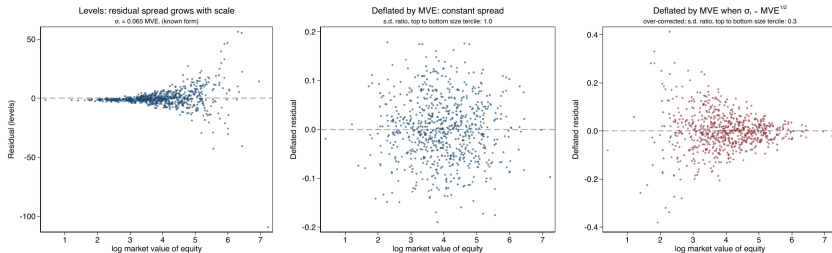
$$\frac{\text{Var}_{\text{true}}(b)}{\text{Var}_{\text{conv}}(b)} = 1 + \frac{\text{Cov}_n(\tilde{x}_i^2, \sigma_i^2)}{\tilde{x}^2 \bar{\sigma}^2}$$

$\sigma_i^2$  constant: ratio 1    ·     $\sigma_i^2$  rising in  $\tilde{x}_i^2$ :  $t$  too large    ·    falling:  $t$  too small

- OLS remains unbiased; only the reported standard error is wrong, by the covariance between the squared regressor and the error variance.

# In levels the residual spread tracks scale

Deflation removes heteroskedasticity only when its form matches the deflator  
Simulated data (seed 1999),  $N = 740$ , calibrated to ABK Table 2; the same draws in all three panels



$\sigma_i = 0.065 z_{\text{med}}^{(2-\gamma)/2} MVE_i^{\gamma/2}$  (s.d.  $0.065 MVE$  at the median firm for every  $\gamma$ );  $\gamma = 2$  in the first two panels,  $\gamma = 1$  in the third.

- Deflation flattens the fan only when the error standard deviation is proportional to the deflator; a smaller exponent leaves a fan that narrows with size, a larger one a fan that still widens.

## One draw: conventional s.e. under half the truth

```
gen z = exp(4+rnormal()) // MVE
qui sum z, detail
scalar zm = r(p50)
gen r = rgamma(.35,.314)
replace r = r*(z/zm)^-.25
gen x = z*r // REV
gen u = .065*z*rnormal()
gen y = .02*z + .07*x + u
reg y x z // levels
reg y x z, vce(robust) // White
gen yd = y/z
gen xd = x/z
gen iz = 1/z
reg yd xd iz // deflated
```

| Regression      | Standard error            | $\hat{\beta}$ | s.e. (t)           |
|-----------------|---------------------------|---------------|--------------------|
| Levels          | conventional $s^2/S_{xx}$ | 0.014         | <b>0.022 (0.7)</b> |
| Levels          | White (1980), HC1         | 0.014         | 0.031 (0.5)        |
| Levels          | White, HC3                | 0.014         | 0.033 (0.4)        |
| Levels          | true, $\sigma_i$ known    | 0.014         | <b>0.055</b>       |
| Deflated by $z$ | conventional              | 0.085         | <b>0.013 (6.5)</b> |
| Deflated by $z$ | White (1980), HC1         | 0.085         | 0.013 (6.7)        |

True  $\beta = 0.07$ ;  $N = 740$ , seed 1999. Identity ( $\sigma_i$  known): 6.16; realized  $(0.055/0.022)^2 \approx 6.3$  since leverage biases  $s^2$  down.

- The conventional levels s.e. understates the truth sixfold in variance; White recovers part of the gap; after deflation the two standard errors coincide.

## White replaces $\sigma_i^2$ with $\hat{u}_i^2$ , no form needed

$$\widehat{\text{Var}}_{\text{HCO}}(\hat{\beta}) = (X'X)^{-1} \left( \sum_i \hat{u}_i^2 x_i x_i' \right) (X'X)^{-1} \quad \text{scalar form: } \frac{\sum_i \tilde{x}_i^2 \hat{u}_i^2}{S_{xx}^2}$$

| Version | Adjustment                                         | Stata       |
|---------|----------------------------------------------------|-------------|
| HC0     | none (White 1980)                                  | vce(hc0)    |
| HC1     | multiply by $n/(n-k)$                              | vce(robust) |
| HC3     | $\hat{u}_i^2/(1-h_{ii})^2$ , $h_{ii}$ the leverage | vce(hc3)    |

Consistent, with independent observations, under any pattern of  $\sigma_i^2$ ; log: vce(robust) minus hand-computed HC1 = -0.00000000.

Caveat: in levels with lognormal scale, HC1 rejects 7 to 14 percent of true nulls at  $N = 740$ , HC3 3 to 7; after deflation, as in ABK, HC1 is correctly sized.

- White's estimator is consistent under any form but not unbiased; under leverage from a skewed scale variable, HC1 understates the variance and HC3 largely restores nominal size.

## Deflation is WLS with weight $1/MVE^2$

Levels model with scale  $z$  (here MVE), true constant  $a = 0$ ; divide every term by  $z_i$ :

$$y_i = a + \alpha z_i + \beta x_i + u_i \implies \frac{y_i}{z_i} = \alpha + a \frac{1}{z_i} + \beta \frac{x_i}{z_i} + \frac{u_i}{z_i}$$

If  $\sigma_i^2 = \sigma^2 z_i^\gamma$ :  $\text{Var}(u_i/z_i) = \sigma^2 z_i^{\gamma-2}$ , constant if and only if  $\gamma = 2$ .

OLS on the deflated variables minimizes  $\sum_i (e_i/z_i)^2 = \sum_i w_i e_i^2$ ,  $w_i = 1/z_i^2$ : weighted least squares, regress  $y$   $x$   $z$  [ $aw = 1/z^2$ ]. Log: deflated and WLS slope, s.e., and intercept agree to  $10^{-8}$ .

- ABK's year intercepts are coefficients on market value in a levels model, and the deflated error is homoskedastic only if the error standard deviation is exactly proportional to market value.

## Each conventional test is right for one form

| Form of $\sigma_i^2$          | $\gamma$ | Levels OLS  |      |      | Deflated by $z$ |             | s.d. ratio  |
|-------------------------------|----------|-------------|------|------|-----------------|-------------|-------------|
|                               |          | conv.       | HC1  | HC3  | conv.           | HC1         |             |
| $\sigma^2$                    | 0        | <b>0.05</b> | 0.07 | 0.05 | <b>0.19</b>     | 0.05        | 6.51        |
| $\sigma^2 z_i$                | 1        | 0.29        | 0.11 | 0.07 | <b>0.11</b>     | 0.05        | 1.07        |
| $\sigma^2 z_i^2$              | 2        | <b>0.47</b> | 0.11 | 0.05 | <b>0.05</b>     | 0.06        | <b>0.21</b> |
| $\sigma^2 z_i^3$              | 3        | 0.54        | 0.11 | 0.05 | <b>0.02</b>     | 0.05        | 0.05        |
| $\sigma^2 z_i^2 (1 + 2D_i)^2$ | unknown  | 0.57        | 0.14 | 0.07 | <b>0.12</b>     | <b>0.07</b> | 0.23        |

Rejection rates of the true null  $\beta = 0.07$  at 5 percent; 1,000 samples of  $N = 740$  per row; s.d. ratio: sampling s.d. of the deflated over the levels slope;  $D_i = 1$  above the median ratio.

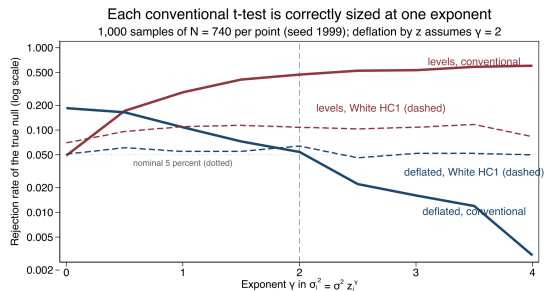
- The conventional levels test is sized correctly only at  $\gamma = 0$  and the conventional deflated test only at  $\gamma = 2$ ; White after deflation stays within two points of nominal in every design.

# Mis-weighting errs as the identity predicts

Levels:  $\sigma_i^2$  and  $\tilde{x}_i^2$  both rise with size, so the covariance term grows with  $\gamma$ .

Deflated: the error variance is  $\sigma^2 z_i^{\gamma-2}$  and small firms have larger ratios; the covariance is positive below  $\gamma = 2$ , negative above.

- A deflator with the wrong exponent misstates the standard error in the direction the covariance identity gives.

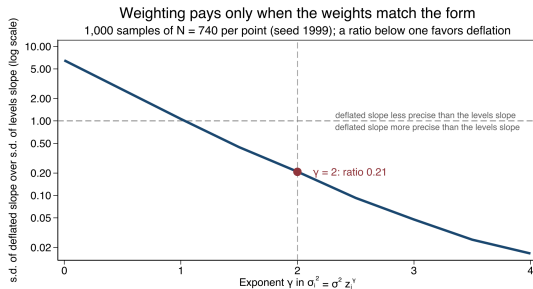


# Weights pay only when they match the form

With the true weights WLS is best linear unbiased (Aitken 1935); with the wrong weights it can be far less precise than OLS.

s.d. ratio: 6.51 at  $\gamma = 0$ ; 1.07 at  $\gamma = 1$ ; 0.21 at  $\gamma = 2$ ; 0.02 at  $\gamma = 4$ .

- Deflation multiplies the slope's sampling s.d. by 6.5 under homoskedastic levels errors and divides it by nearly five under errors proportional to MVE.

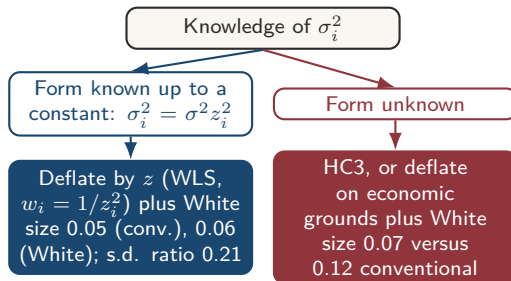


## The choice follows what is known about $\sigma_i^2$

Known form: weight by it (deflate); add White at little cost.

Unknown form: HC3 under leverage, or deflate on economic grounds plus White.

- Deflation imposes one form on the error variance; a heteroskedasticity-consistent covariance protects inference when that form is wrong.



## ABK's $t$ -statistics rest on three layers

| Layer              | What it does              | What it assumes                  | Stated        |
|--------------------|---------------------------|----------------------------------|---------------|
| Deflation by MVE   | WLS, weight $1/MVE^2$     | pound errors proportional to MVE | p. 160; fn 12 |
| White (1980)       | consistent variance       | any remaining form; independence | p. 162        |
| Robust reweighting | a second, data-chosen WLS | weights from absolute residuals  | fn 17         |

Footnote 17: after a Cook's D screen, weights “are calculated based on absolute residuals” and the estimation is repeated on the weighted observations (p. 162).

- The published  $t$ -statistics appear to be White standard errors of a residual-reweighted, deflated, year-effects regression; the text does not say on which pass they were computed.

## ■ Table 3 supports the sign more than the $t$

Every layer leaves the REV slope significantly positive at each horizon (Table 3, p. 163: 0.07, 0.05, 0.12), undeflated (fn 12) and under OLS (fn 17) as well.

*“Our inferences are unaltered if we estimate the equations using ordinary least squares and if we remove observations with an  $R$ -student statistic greater than three in absolute value, rather than use the robust regression procedure.”*

Aboody, Barth, and Kasznik (1999, p. 162, fn 17)

A skeptic would grant the sign and question only the size of  $t = 6.96$  (Table 3, p. 163).

- The corrections protect the sign of the revaluation coefficient; the size of its  $t$ -statistic rests on a weighting scheme the paper only partly discloses.

# Summary

- Heteroskedasticity leaves the OLS slope unbiased and misstates its standard error by the covariance between the squared regressor and the error variance; where both track firm size, conventional  $t$ -statistics appear too large.
- White's (1980) estimator is consistent under any form of heteroskedasticity and is the correction of choice when the form is unknown; with a skewed scale variable and a few hundred firms, HC1 is biased downward and HC3 is advisable.
- Deflation is weighted least squares under the assumption that the error standard deviation is proportional to the deflator; it gains precision when that holds and loses it, with mis-sized tests, otherwise, so a deflator chosen on economic grounds should be paired with White standard errors, as in ABK.

Aboody, Barth, and Kasznik (1999) · White (1980) · Barth and Kallapur (1996) ·  
[github.com/sbb151/acctg597a-research-design](https://github.com/sbb151/acctg597a-research-design)